

Geriatric Trauma with Head Injury



Section I: Scenario Demographics

Scenario Title:	Geriatric Trauma with Head Injury		
Date of Development:	(20/04/2018)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Professor Victoria Brazil, Dr Nemat Alsaba, Dr Rebecca Shaw
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To practice focused assessment and management of an elderly patient with a severe traumatic brain injury, complicated by anticoagulation and elevated ICP
CRM Objectives:	1. Use a team-based approach for assessment of an elderly trauma patient 2. Manage multiple simultaneous medical needs of a severely injured patient 3. Ensure the family are well supported and well informed about the patient
Medical Objectives:	1. Assess an elderly trauma patient, accounting for geriatric trauma pathophysiology 2. Perform a safe intubation in a geriatric trauma patient with a severe brain injury 3. Manage elevated ICP in the setting of trauma 4. Reverse a NOAC in the setting of trauma

Case Summary: Brief Summary of Case Progression and Major Events

An 81-year old man falls down the stairs at home. He is initially asymptomatic but his level of consciousness declines and he starts to show signs of raised ICP. Providers must recognize and treat this, as well as reverse his anticoagulation, provide neuroprotective RSI and safely transport to the CT scanner. Providers must then talk with the patient's wife, to provide information on his condition and prognosis and discuss the patient's goals of care.

References

<https://lifeinthefastlane.com/what-the-elderly-should-say/>
<http://www.emdocs.net/geriatric-trauma-medical-illness-pearls-pitfalls/>
https://www.health.qld.gov.au/_data/assets/pdf_file/0026/147662/qh-gdl-950.pdf
<https://canadiem.org/crackcast-e039-geriatric-trauma/>

Geriatric Trauma with Head Injury



Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case

Bedside nurse to read to team leader: "EMS just off-loaded an elderly male to the resuscitation bay. He had a fall down the stairs and sustained a head injury. He was GCS 15 and hemodynamically stable when they picked him up, so they didn't activate the trauma team, but he has deteriorated during transport. He has got an obvious large, boggy scalp hematoma over the left parietal region. I am worried because he's getting restless and won't follow commands."

B. Scenario Cast & Realism

Patient:	☒ Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☐ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates Brief Description of Role

ED RN	To give initial hand over as described above, assist with patient care at the bedside, and alert team to changes in status.
Patients wife	You were present when your husband fell down the stairs and are very concerned about what has happened to him. He has a few medical problems and is on a few medications but you both manage well at home. You are unaware how unwell your husband is or that he has been intubated. The participant should inform you that your husband likely has a bleed on his brain and has been intubated. He is awaiting a CT scan of his head and a neurosurgical and ICU review. They should inform you that the prognosis in this case may be poor but that your husband is currently being actively managed. If they ask you what his wishes are regarding resuscitation inform them that you have never discussed this. If they use medical jargon rather than simple terms to explain his management or prognosis ask for further clarification. You should be upset but calm during the discussion.

C. Required Monitors

☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☒ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:



Geriatric Trauma with Head Injury



D. Required Equipment

☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☒ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage

Boggy hematoma on the left parietal area.

F. Approximate Timing

Set-Up:	10 min	Scenario:	20 min	Debriefing:	20 min
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Geriatric Trauma with Head Injury



Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Clarence Worthington		Age: 81		Weight: 70 KG	
Gender: ☒ M ☐ F		Code Status: Unknown			
Chief Complaint: Head injury					
History of Presenting Illness: Fell down the stairs at home hitting his head. Initially he was fine but then started to become agitated en route to hospital.					
Past Medical History:	Ischemic Heart Disease		Medications:	Rivaroxaban	
	HTN			Metoprolol	
	AF			Furosemide	
	COPD			Unknown puffers	
Allergies: Nil known allergies					
Social History: Lives independently at home with his wife, Jean					
Family History: Nil					
Review of Systems:	CNS:	Agitated, cannot provide history			
	HEENT:				
	CVS:				
	RESP:				
	GI:				
	GU:				
	MSK:		INT:		
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☐ Monitor On, no data displayed		☒ Monitor on Standard Display	
HR: 85 /min	BP: 115/60	RR: 16/min	O ₂ SAT: 94 %		
Rhythm: A fib	T:36.1°C	Glucose:5.0 mmol/L	GCS: 13 (E4 V4 M5)		
General Status: Agitated, confused.					
CNS:	Confused. Not following commands. Moving all 4 limbs				
HEENT:	Boggy hematoma left parietal area.				
CVS:	Irregularly irregular. No murmurs.				
RESP:	Clear breath sounds bilaterally.				
ABDO:	Soft, non-tender.				
GU:	Nil				
MSK:	Nil	SKIN:	Nil		

Geriatric Trauma with Head Injury



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers

Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: A Fib HR: 85/min BP: 115/60 RR: 16/min O ₂ SAT: 94% T: 36.1°C	Agitated. Not oriented. Talkative. Not making sense. Pupils equal and reactive. (GCS 14)	<u>Learner Actions</u> - ☐ Primary Survey - ☐ IV access - ☐ Trauma labs - ☐ FAST - ☐ CXR + Pelvic XR - ☐ Call wife for collateral history (unavailable, on her way to the hospital) - ☐ Ask for med list (given)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Increase O ₂ Sat to 98% if supplemental O ₂ given. <u>Triggers</u> <i>For progression to next state</i> - All actions complete or after 3 mins → 2. Deterioration
2. Deterioration Rhythm: A Fib HR: 85/min BP: 115/60 RR: 16/min O ₂ SAT: 98% T: 36.1°C	Patient much less agitated. Drowsy. GCS 9 (E1V3M5). Pupils equal and reactive.	<u>Learner Actions</u> - ☐ Consider CT head - ☐ Consider NOAC reversal - ☐ Consider intubation - ☐ Complete primary survey, including log roll	<u>Modifiers</u> - CXR/PXR: negative <u>Triggers</u> - All actions complete or after 3 mins → 3. Raised ICP
3. Raised ICP Rhythm: A Fib HR: 50/min BP: 190/110 RR: 12/min O ₂ SAT: 98% T: 36.1°C	Patient vomits. GCS 5 (E1V1M3) Left pupil dilated.	<u>Learner Actions</u> - ☐ Treat increased ICP (hypertonic saline or mannitol) - ☐ Raise head of the bed - ☐ Decision to intubate - ☐ Neuroprotective RSI - ☐ NOAC reversal (if not yet done)	<u>Modifiers</u> - Hypertonic saline or mannitol given → BP 170/90, HR 60 - If not moving to intubate, patient to vomit again and drop O ₂ sats to 88% <u>Triggers</u> - Successful intubation → 4. Post-intubation
4. Post-Intubation Rhythm: A Fib HR: 60/min BP: 150/90 RR: 12 (vented) O ₂ SAT: 100% T: 36.1°C	Patient intubated and ventilated	<u>Learner Actions</u> - ☐ Post-intubation sedation - ☐ Post-intubation CXR - ☐ Complete ICP management - ☐ Complete NOAC reversal - ☐ Arrange CT scan - ☐ Call ICU, neurosurgery	<u>Modifiers</u> <u>Triggers</u> - All actions complete → Patient goes to CT and learner notified that wife has arrived (or is on the phone) → 5. Discussion with wife
5. Discussion with wife	Patient stable in CT	<u>Learner Actions</u> - ☐ Compassionate discussion with patient's wife - ☐ Determine patient's goals of care - ☐ Involve social worker	<u>Modifiers</u> <u>Triggers</u> - Wife informed of condition, management and likely prognosis → End of case

Geriatric Trauma with Head Injury



Section VII: Supporting Documents, Laboratory Results, & Multimedia

No lab results given during case.

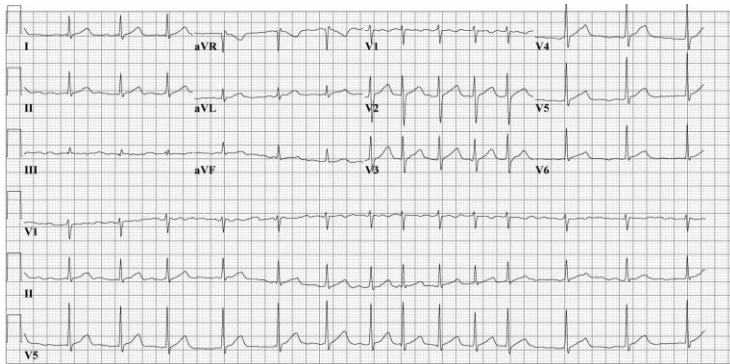
Images (ECGs, CXRs, etc.)



Case courtesy of Dr Jeremy Jones, Radiopaedia.org, rID: 6410



Case courtesy of Dr Jeremy Jones, Radiopaedia.org, rID: 28928



https://en.ecgpedia.org/index.php?title=Atrial_Fibrillation

Geriatric Trauma with Head Injury



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
Medical 1. What are the differences between elderly trauma patients and younger adults? 2. What were your management priorities in this patient? 3. What strategies did you use to safely intubate this head injured patient? How did this inform your choice of RSI medications? 4. What neuroprotective measures did you implement in this case? 5. How did you reverse the patient's rivaroxaban?	
CRM 1. Did you work effectively as a team to assess this patient? How did you organize the team to manage this patient? 2. How did you decide which tasks to prioritize in this patient? What strategies do you use when managing patients with multiple competing priorities at once? 3. How did you feel discussing the patient with his wife? What factors do you consider when preparing for this discussion?	
Key Moments	
1. Recognition of severe head injury.	
2. Institution of neuroprotective measures and reversal of NOAC.	
3. Decision to intubate.	
4. Discussion with family.	